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Combined Environmental Impacts and Toxicological Interactions of Per- and Polyfluoroalkyl Substances (PFAS) and Microplastics (MPs)

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Abstract

Pervasive microplastics (MPs) and per- and polyfluoroalkyl substances (PFAS) frequently co-occur across aquatic and terrestrial environments due to shared sources, transport pathways, and persistence, yet their interaction-driven effects on environmental fate, bioavailability, and toxicity remain incompletely resolved. This review critically synthesizes current knowledge on the environmental co-occurrence of MPs and PFAS, the physicochemical mechanisms governing their interactions, and the resulting ecological and toxicological consequences across aquatic, terrestrial, and biological systems. Emphasis is placed on sorption and desorption processes; environmental modifiers such as pH, salinity, dissolved organic matter (DOM), and aging; and biological responses under combined exposure scenarios. Across laboratory and field studies, MPs–PFAS co-exposure is frequently associated with altered PFAS partitioning and enhanced organismal uptake, with reported bioaccumulation increases of up to ~2.5-fold relative to PFAS-only exposures. These changes are often accompanied by amplified oxidative stress, immune dysregulation, metabolic disturbance, and reproductive impairment, particularly in aquatic invertebrates and early life stages of fish. Evidence further indicates that the magnitude and direction of combined effects depend on polymer type, particle size, surface aging, and biological context, underscoring the highly system-specific nature of MPs–PFAS interactions. By integrating findings from environmental monitoring, laboratory toxicology, and mechanistic and modeling studies, this review identifies key knowledge gaps related to nanoplastics detection, environmentally realistic exposure conditions, sorption reversibility, and mixture toxicity assessment. Collectively, these insights highlight limitations in current single-contaminant risk frameworks and underscore the importance of incorporating MPs-mediated PFAS transport and bioavailability into exposure assessment and regulatory evaluation.



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Keywords: environmental impacts; toxicological interactions; per- and polyfluoroalkyl substances (PFAS); microplastics (MPs); regulations and policies; co-occurrence

1. Introduction

Aquatic systems are often the first environmental bodies to accumulate emerging contaminants. Microplastics (MPs), similar to per- and polyfluoroalkyl substances (PFAS) and endocrine-disrupting chemicals (EDCs), are typically transported from land bodies, like road surfaces or soil, into aquatic environments [1]. While extensive research has

demonstrated how MPs adsorb hydrophobic organic contaminants (HOCs), detailed investigations into their interactions with PFAS and EDCs are limited. The joint presence of these pollutants is a growing concern as simultaneous exposure is known to amplify adverse effects, as a study involving freshwater organisms observed higher toxic effects of MPs combined with PFAS [2]. Evidence indicates that such co-exposure may lead to combined toxic responses, such as synergistic effects in aquatic organisms [3]. Notably, current research has yet to fully examine the complex interactions involving all three pollutant groups such as MPs, PFAS, and EDCs together.

Despite the relevance of multiple co-occurring contaminants, this review places particular emphasis on PFAS because they represent a uniquely persistent and environmentally-mobile class of pollutants when considered alongside MPs. Unlike many other organic contaminants, PFAS strongly resist chemical, biological, and photolytic degradation, resulting in widespread accumulation in aquatic systems, wildlife, and human tissues. PFAS also exhibit strong and well-documented sorption to polymer surfaces, allowing MPs to act not only as passive carriers but as active vectors that enhance PFAS transport, bioavailability, and internal exposure. Given their global ubiquity, demonstrated biological impacts, and increasing regulatory concern, PFAS provide a particularly relevant focus for understanding MPs-mediated contaminant interactions.

Multiple studies have found that when PFAS attach to or coat the surfaces of MPs, they can act as an additional source of environmental contamination [4–6]. Both MPs and PFAS tend to persist and accumulate over long time periods because they are highly resistant to degradation, resulting in chronic exposure risks [7]. MPs have been found to enhance transport of PFAS compounds through soil in some cases [8]. Once contaminated by sorbed HOCs, these MPs may be consumed by aquatic organisms, allowing PFAS to move through food webs, posing ecological as well as human health hazards [9,10]. Moreover, the gradual release of PFAS from MPs surfaces further amplifies their distribution in the environment, intensifying major concerns about long-term impacts. Considering the persistent and ubiquitous nature of MPs and PFAS in the environment, understanding their interactions and the consequences for ecosystems and human health is of paramount importance [11].

MPs originate from multiple sources, including the fragmentation of larger plastic debris, abrasion of synthetic textiles, tire wear particles, and industrial emissions. Once released, MPs act as environmental sinks for co-existing contaminants due to their high surface area and chemical heterogeneity [2,12]. Recent toxicological studies demonstrate that MPs–PFAS co-exposure can enhance bioavailability and toxicity beyond that observed for individual contaminants, underscoring the need for focused synthesis of these interactions [13,14]. Figure 1 integrates the major environmental sources and co-occurrence of MPs and PFAS with molecular-scale physicochemical interaction mechanisms; key environmental controls on sorption; and resulting exposure, uptake, and ecotoxicological pathways that frame the mechanisms discussed in Section 2 [11].

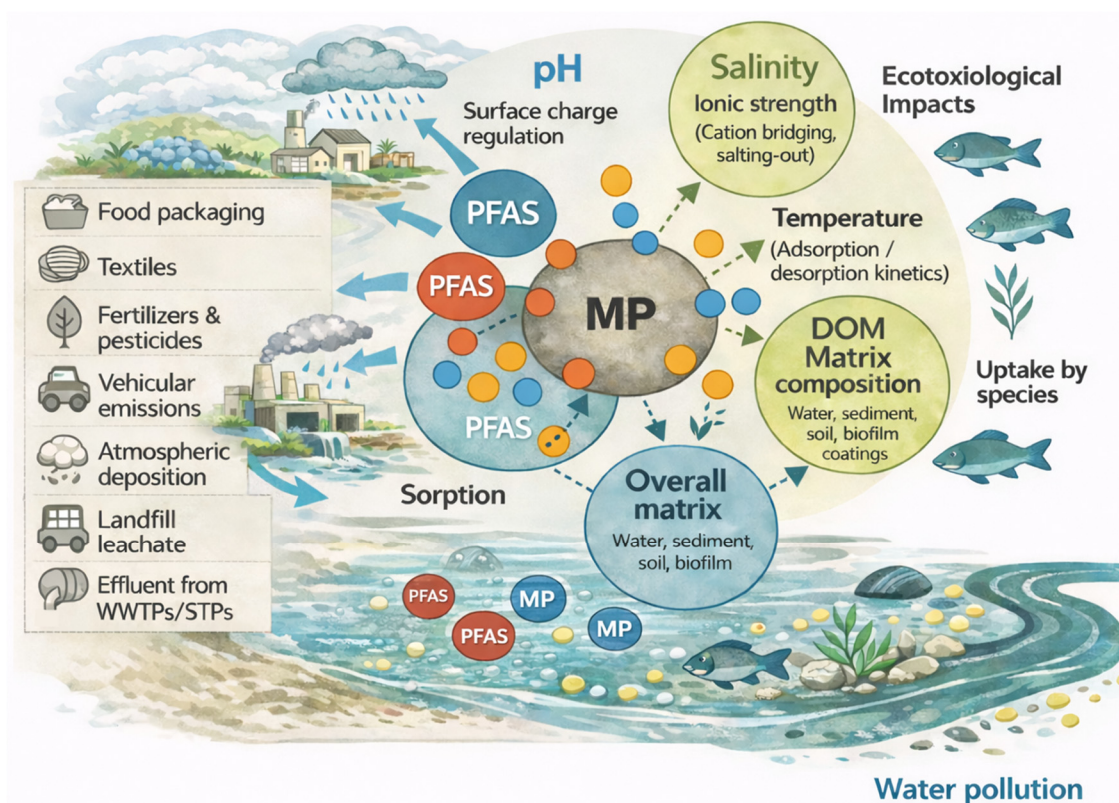


Figure 1. Integrated framework showing environmental sources, co-occurrence, and exposure pathways of MPs and PFAS, together with molecular-scale interaction mechanisms governing PFAS sorption to MPs. MPs are represented by the large central grey particle, while PFAS are depicted as smaller colored spheres distributed in the surrounding aqueous matrix and at the MPs surface. PFAS–MPs interactions include hydrophobic association, electrostatic interactions, pore filling, competitive binding, and dissolved organic matter (DOM) mediated adsorption. Sorption behavior is further modulated by environmental factors including pH, salinity, temperature, and overall matrix composition (water, sediment, soil, and biofilm).

2. Literature Search and Methodology

This review is based on a structured literature survey designed to synthesize current knowledge on the environmental co-occurrence, interaction mechanisms, and toxicological effects of MPs and PFAS. Literature searches were conducted using Scopus, Web of Science, and Google Scholar. Searches combined keywords including “microplastics,” “MPs,” “nanoplastics,” “NPs,” “PFAS,” “per- and polyfluoroalkyl substances,” “sorption,” “co-exposure,” “MPs–PFAS interactions,” “mixture toxicity,” “carrier effects,” and “vector behavior.” Priority was given to peer-reviewed articles published between 2015 and 2025, with particular emphasis on studies from the last five years addressing physicochemical interactions, environmental modifiers, and toxicological outcomes under combined MPs–PFAS exposure. Additional references were identified through backward citation tracking of key review articles.

3. Co-Existence of MPs and PFAS in the Environment

MPs and PFAS frequently co-occur in environmental systems due to shared sources, overlapping transport pathways, and comparable persistence characteristics. Both contaminant classes are released through wastewater treatment plant (WWTP) effluent, urban stormwater runoff, landfill leachate, and industrial discharges, as well as from consumer products such as food packaging, synthetic textiles, surface coatings, and non-stick materials [1,2,15]. As a result, MPs and PFAS are commonly detected together in surface

waters, sediments, soils, and biota, creating chronic co-exposure scenarios across aquatic and terrestrial environments [12,16].

The environmental overlap of MPs and PFAS is particularly concerning because both are highly resistant to degradation and persist over long temporal scales. PFAS are extremely mobile, chemically stable, and bioaccumulative, while MPs provide abundant and reactive surfaces capable of interacting with surrounding contaminants. Field observations increasingly demonstrate that environmental exposure rarely occurs in isolation; rather, organisms encounter complex mixtures in which MPs and PFAS are simultaneously present [2,15,17].

Although MPs can associate with a wide range of co-pollutants, PFAS represent a particularly relevant contaminant class due to their exceptional persistence, strong affinity for polymer surfaces, global distribution, and well-documented ecological and human health effects. Sorption of PFAS onto MPs surfaces enables MPs to act not only as passive carriers but also as active vectors that alter PFAS transport, bioavailability, and internal exposure pathways [5,6,11]. The frequent environmental co-existence of MPs and PFAS therefore creates conditions conducive to physicochemical interaction and combined biological effects, which form the basis for the mechanistic and toxicological discussions presented in subsequent sections of this review.

3.1. Characteristics, Sources, and Impacts of MPs

With the environmental co-existence of MPs and PFAS established in Section 3, the following section summarizes the characteristics, sources, and biological impacts of MPs relevant to their interaction with co-occurring contaminants. MPs are defined as plastic fragments smaller than 5 mm in size and can extend into the sub-micron range, whereas nanoplastics (NPs) are generally defined as plastic particles with one or more dimensions below ~100 nm. References to ~1 μm particles therefore correspond to sub-micron MPs rather than true nanoplastics [18–20]. MPs represent not one material but a complex mixture of synthetic polymers—most commonly, polyethylene (PE) and polypropylene (PP), with some polyvinyl chloride (PVC), polyethylene terephthalate (PET), and polystyrene (PS). Since each polymer has its own particular density, flexibility, and surface chemistry, the environmental behavior of plastic materials made from them can be extremely diverse. It is these differences that are important in controlling how particles might float and disperse or settle within marine waters and subsequently interact with other compounds. MPs adopt various physical forms in the environment: fiber (e.g., textiles), fragments (bigger plastics), films, foams, and regular or irregular spheres. Their surface chemistry varies also, where some stay relatively unchanged while others are oxidized or coated with microbial biofilms. The diversity in structure, chemistry, and particle size directly determines how these plastics may move through the water column, accumulate contaminants, and affect marine organisms [21].

In addition to conventional petroleum-based polymers, recent classification frameworks increasingly recognize MPs as particles derived from both conventional and biodegradable polymers. These include materials such as polylactic acid (PLA), polyhydroxyalkanoates (PHAs), cellulose-based fibers, and starch-derived polymers, which are frequently marketed as environmentally sustainable alternatives. However, growing evidence indicates that many biodegradable plastics fragment into microscale and nanoscale particles under environmentally realistic conditions and persist far longer than expected. Studies have shown that these biodegradable polymer fragments can exhibit transport behavior, surface reactivity, and contaminant sorption characteristics comparable to those of conventional plastics, raising concerns about their role in MPs pollution rather than their

mitigation [22]. Consequently, both fossil-derived and biodegradable polymer particles are increasingly considered within the definition of MPs in environmental research.

Environmental aging processes, including ultraviolet (UV) irradiation, pH fluctuations, oxidizing conditions, and hydrodynamic abrasion, progressively modify MPs surfaces by increasing roughness, porosity, and the abundance of oxygen-containing functional groups. These transformations enhance interactions with DOM, including the release and formation of MPs-DOM, which actively participates in aquatic biogeochemical processes [23]. DOM coatings alter MPs surface charge, polarity, and sorption capacity, thereby enhancing the adsorption of PFAS through hydrophobic interactions, electrostatic attraction, and cation bridging mechanisms [23,24]. DOM can also modify PFAS partitioning behavior and mobility, influencing MPs-mediated cotransport and environmental fate [24]. In addition, MPs inherently contain chemical additives such as plasticizers, antioxidants, and stabilizers, which may be mobilized and released as surface weathering progresses, contributing to secondary chemical pollution and mixed-contaminant exposure scenarios [25,26].

NPs present substantial methodological challenges for environmental monitoring due to their extremely small size, low mass, and heterogeneous composition. Unlike MPs, NPs cannot be reliably captured using conventional filtration, visual inspection, or density separation, making representative environmental sampling difficult. Current approaches rely on indirect or multi-step isolation procedures such as ultrafiltration, asymmetric flow field-flow fractionation (AF4), ultracentrifugation, or cloud-point extraction, often followed by advanced analytical detection [27]. Identification and characterization typically employ Raman spectroscopy, Fourier-transform infrared spectroscopy (FTIR), thermal desorption–gas chromatography–mass spectrometry (TD-GC/MS), or pyrolysis-GC/MS; however, each technique is limited by detection thresholds, spectral overlap, particle aggregation, and interference from natural organic matter [28,29]. As a result, quantitative assessment of environmental matrices remains highly uncertain, and standardized protocols for sampling, identification, and reporting are still lacking, underscoring a critical gap in NP research.

Primary MPs are intentionally manufactured at microscopic sizes prior to direct release into the environment. Common sources include cosmetic microbeads, household items, and microfibers shed from synthetic clothing during washing [30]. In contrast, secondary MPs result from the degradation of larger plastic items through erosion, friction, and environmental wear. Compared with primary MPs, they are more irregular in shape, size, and composition and are more prevalent in marine ecosystems [30].

MPs act as vectors for co-pollutants. MPs can sorb and carry organic contaminants and PFAS. Sorption is shaped by hydrophobic, electrostatic, and other interactions (which are in turn shaped by particle aging, pH, salinity, and DOM). These processes influence contaminant retention, mobility, and bioavailability and are now included in mechanistic reviews and modeling frameworks [31]. In natural waters, MPs rapidly acquire biofilms and organic coatings that alter surface charge/energy, volumetric mass density (potentially promoting sinking/aggregation), and sorption behavior. These alterations, referred to as biofilm and eco-corona effects, change where particles travel and what they deliver. WHO's synthesis for drinking water has framed MPs-related drinking water hazards as (i) the physical particle, (ii) released chemicals, and (iii) microbes carried in biofilms [32].

MPs have emerged as a widespread pollutant with measurable biological impacts across multiple ecosystems. In ocean ecosystems, filter feeders like mussels are especially prone to ingesting MPs, which tend to accumulate in their gills and digestive tracts. This buildup can lead to a cascade of physiological issues, including inflammation of tissues, destabilization of lysosomes, weakened immune responses, and disruptions in how energy is processed [33–35]. Crustaceans such as shrimp and copepods also exhibit signs of stress when exposed to MPs. Their feeding becomes less efficient, growth slows, and they often

experience oxidative damage. Behavioral shifts, like changes in swimming patterns, have also been observed [36,37]. Fish appear to be particularly sensitive to MPs contamination. Studies have found MPs lodged in their gills, intestines, and liver. These particles can block the digestive tract, interfere with nutrient uptake, and cause structural damage to liver tissue. In some cases, MPs have been linked to hormonal imbalances and reduced fertility [38,39]. The cumulative impact of these effects raises serious concerns about the long-term health of marine species and the ecosystems they support.

The mechanisms underlying MPs toxicity are both physical and chemical. Physically, MPs may cause tissue abrasion, intestinal obstruction, and reduced feeding efficiency. Chemically, MPs act as vectors for leached plastic additives such as phthalates and bisphenols and can adsorb environmental pollutants like PFAS and heavy metals, enhancing their bioavailability and toxicity [3]. Biological responses commonly observed include oxidative stress, immune modulation, reproductive impairment, and genotoxicity, with species-specific susceptibility depending on physiology, exposure route, and particle characteristics [40]. Together, these findings support the classification of MPs as biologically active contaminants capable of affecting the health of individual organisms, populations, and ecosystems.

The environmental transport and fate of MPs involve long-range transport, physico-chemical transformation, and accumulation in sediments and biota, which are dependent on numerous physical and chemical properties. MPs act as vectors for chemical pollutants, have long environmental residence times, and are present in virtually every ecosystem; comprehensive monitoring, management, and regulatory strategies are increasingly imperative [41]. Despite extensive documentation of MPs occurrence and toxicity, substantial knowledge gaps remain regarding environmentally realistic exposure conditions and interaction-driven effects [40]. Most experimental studies rely on pristine particles at concentrations exceeding those typically observed in nature, limiting ecological relevance and risk extrapolation [42]. Furthermore, the influence of particle aging, biofilm formation, and polymer-specific behavior on sorption capacity and toxicity remains inconsistently addressed, despite strong evidence that these processes substantially alter surface chemistry, contaminant binding, and biological responses [23,40]. These limitations are particularly consequential for mixture scenarios involving PFAS, where sorption kinetics, reversible desorption, and biological bioaccessibility can significantly alter internal dose, tissue distribution, and toxicological outcomes relative to dissolved exposure alone [11,43]. Addressing these gaps requires standardized exposure frameworks, integration of environmentally realistic aging and biofilm processes, and explicit consideration of MPs as dynamic rather than inert stressors in mixture-toxicity assessments [40–42].

In mammals, laboratory studies have shown that MPs ingested by rodents can translocate beyond the gastrointestinal tract, accumulating in the liver, kidneys, and even brain tissue. Such accumulation often depends on the MPs particle size. This systemic distribution has been linked to oxidative stress, inflammation, metabolic disruption, and gut microbiome imbalance [43–46]. There is strong evidence that humans are also exposed to MPs primarily through direct ingestion (e.g., seafood, drinking water, food packaging, etc.) and inhalation of airborne MPs fibers. Human stool, placenta, and lung tissue have been found to contain MPs, demonstrating the ability of MPs to cross biological barriers [46–48].

In humans, potential health outcomes include respiratory irritation, systemic inflammation, endocrine disruption, and oxidative stress, although the long-term clinical consequences remain uncertain [49]. Experimental studies consistently demonstrate that MPs induce oxidative stress, inflammation, and mitochondrial dysfunction, processes that can lead to apoptosis, immune activation, and impaired barrier integrity in epithelial tissues [49–51]. These mechanisms are directly relevant to chronic disease, as persistent inflammation

and oxidative damage are central to cardiovascular, metabolic, and neurodegenerative disorders [29,40]

MPs have been detected in human blood, confirming systemic exposure and circulation potential [16,52]. MPs and NPs have also been identified in the placenta, lung, liver, and kidney, raising concern about developmental and multi-organ effects [17]. Perhaps most strikingly, a 2024 study in the *New England Journal of Medicine* found MPs embedded in carotid artery plaques, with their presence linked to a fourfold increased risk of myocardial infarction, stroke, or death over a three-year follow-up [53]. These findings suggest MPs may act not only as passive contaminants but also as active contributors to cardiovascular pathology.

Although causality in humans has not been fully established, the convergence of mechanistic evidence with early clinical observations indicated credible public health risk. MPs are not inert: they carry toxic additives and co-pollutants such as PFAS and PAHs, alter immune and endocrine function, and disrupt microbiome balance [18]. NPs have been implicated in increased cancer rates due to enhanced transport of adsorbed toxins [54]. Future research must quantify exposure thresholds; clarify vulnerable populations (e.g., children and patients with preexisting conditions); and assess the cumulative impact of chronic, low-level exposures. Until then, MPs should be considered emerging human toxicants with potentially significant implications for public health.

3.2. Environmental Occurrence and Toxicological Profile of PFAS

Building on the co-existence framework outlined above, this section reviews the environmental occurrence, chemical properties, and toxicological profile of PFAS relevant to their joint presence with MPs. Often referred to as forever chemicals due to their exceptional resistance to degradation, PFAS are detectable everywhere from Arctic ice to urban stormwater. Although widespread awareness by the public is fairly recent, these compounds have been produced for more than seventy years. During this time, PFAS have been incorporated into more than 200 industrial and commercial applications, representing over 1400 distinct chemicals. Their uses range from food packaging and stain-resistant textiles to firefighting foams, with extensive application in non-stick cookware and numerous other industrial and consumer products since their introduction in the 1940s. The expansion of PFAS applications stems from the development of a vast and chemically varied group of synthetic substances comprising over 1400 distinct chemicals that all share a backbone of highly durable perfluorinated carbon bonds [55].

At each stage of their life cycle, such as manufacture, consumer use, and disposal, PFAS can escape into the environment. Over time, they tend to accumulate in oceans and marine sediments, although uptake by plants and animals is also common [56,57]. Numerous studies have documented bioaccumulation in both aquatic and terrestrial organisms [12,15,50,58,59]. While large-scale monitoring programs are now underway, the long-term health implications for ecosystems and humans, especially regarding newer replacement PFAS compounds, remain uncertain. The enormous number of compounds in this class means that environmental detection and risk characterization often lag behind the continual emergence of next-generation PFAS [55].

PFAS represent a class of man-made compounds designed for their strong resistance to water, oil, and staining. Their distinctive structure, comprising carbon chains fully or partially bonded with fluorine atoms, gives them exceptional stability, preventing rapid breakdown in natural systems [60]. Because of this, PFAS persist in the environment for long durations as they are largely unaffected by chemical, biological, or photolytic degradation [61]. The presence of sulfonate or carboxylate functional groups provides a polar head, in contrast to their highly nonpolar tail, which contributes to increased solubility

and facilitates their movement and distribution across environmental compartments [62,63]. Research consistently shows that most PFAS resist complete mineralization in natural settings, making them stand out as some of the most enduring organic contaminants known [64].

The occurrence of multiple PFAS compounds has been shown to heighten the potential for bioaccumulation and adverse effects in aquatic species. Studies consistently report that long-chain PFAS contribute to physiological stress across a range of organisms, including fish, daphnids, and several aquatic plant species [20]. Exposure to perfluorooctanoic acid (PFOA) in microalgae such as *Thalassiosira pseudonana* resulted in inhibited growth, diminished photosynthetic performance, elevated production of reactive oxygen species (ROS), and enhanced activity of antioxidant defense enzymes [20].

The ecological effects of PFAS show strong parallels to the risks these substances pose in humans. Globally, PFAS contamination has been associated with immune suppression, liver dysfunction, and impaired reproduction in wildlife, for instance, reduced hatching success in bird populations and reproductive disorders in turtles [65,66]. Surveys have detected PFAS in over 600 wildlife species, many of which are already classified as threatened or endangered [67]. This contamination compounds existing pressures such as habitat degradation, overexploitation, and broader pollution. By undermining immune resilience and reproductive capacity, PFAS exposure further diminishes the ability of vulnerable populations to recover and sustain their numbers [65–67].

The risks PFAS pose in humans show strong parallels to the ecological effects of these substances. Epidemiological and experimental studies over the past decade have shown that exposure to PFAS during pregnancy and early childhood can interfere with normal growth and development. Prenatal exposure has been associated with reduced birth weight, delayed physical maturation, and subtle congenital anomalies, consistent with PFAS interference in endocrine and metabolic pathways [57,68,69]. Evidence further indicates that the effects of PFAS exposure extend beyond birth. Children with measurable PFAS body burdens have been reported to display lower cognitive performance, increased prevalence of attention-related disorders such as ADHD, and elevated anxiety or emotional dysregulation [68,70]. Continued exposure through childhood and adolescence has also been linked to early puberty, thyroid disruption, kidney stress, and altered reproductive hormone levels, underscoring the long-term developmental sensitivity to PFAS [69,70].

PFAS function as an EDC has also been correlated with damage to lipid metabolism and obesity [69]. Additionally, the U.S. Environmental Protection Agency (EPA) [71] has provided a list of the potential health risks associated with PFAS, such as reduced ability of the body's immune system to fight infections, including reduced vaccine response; interference with the body's natural hormones; and the increased risk of some cancers including prostate, kidney, and testicular [70].

3.3. Status of Management and Mitigation of MPs Pollution

At present, management and mitigation of MPs pollution is addressed through a fragmented combination of plastic source controls, waste management regulations, and chemical safety frameworks, with limited explicit consideration of MPs as vectors for co-occurring contaminants such as PFAS. In the European Union, recent regulatory advances include restrictions on intentionally-added MPs under the REACH Regulation (EU) 2023/2055, which targets sources such as cosmetic microbeads and infill materials, as well as the adoption of harmonized methodologies for measuring MPs in drinking water under the revised Drinking Water Directive [71]. While these measures significantly advance monitoring and upstream source reduction, they do not explicitly address contaminant-loaded MPs or their role in facilitating secondary exposure pathways.

In the United States, mitigation strategies have largely focused on chemical regulation rather than particulate transport mechanisms. The EPA finalized enforceable drinking water limits for multiple PFAS using a mixture-based Hazard Index approach and, in 2024, designated PFOA and perfluorooctanesulfonic acid (PFOS) as hazardous substances under CERCLA [70,72]. However, no federal framework currently evaluates PFAS sorbed to MPs particles as a distinct or cumulative exposure compartment, and MPs mitigation remains indirect, addressed primarily through solid waste management, wastewater treatment optimization, and emerging state-level monitoring initiatives.

Across jurisdictions, these regulatory developments highlight a persistent gap between the management of plastic particles and chemical contaminants, despite growing evidence that MPs can transport, concentrate, and modulate the bioavailability of PFAS and other persistent pollutants. Recent synthesis studies emphasize that current policies insufficiently capture interaction-driven risks as most regulatory frameworks treat plastics and associated chemicals as independent stressors rather than coupled environmental hazards [11]. Addressing this disconnect will require integrated monitoring, reporting, and risk assessment approaches that explicitly recognize MPs as active contaminant carriers rather than inert debris.

4. Mechanisms of Interaction

The interaction mechanisms between MPs and PFAS described in this section can be conceptually unified using Derjaguin–Landau–Verwey–Overbeek (DLVO)-based and extended DLVO-based (XDLVO) interaction frameworks that integrate van der Waals forces, electrostatic interactions, and non-electrostatic surface forces to describe particle–particle and particle–solute behavior in aqueous systems [73]. Recent environmental studies increasingly apply DLVO/XDLVO concepts to explain how hydrophobic partitioning, electrostatic attraction or repulsion, and surface heterogeneity collectively govern PFAS sorption to polymeric surfaces, with interaction energies strongly modulated by pH, ionic strength, and DOM [12,74]. Application of these interaction energy frameworks provides a mechanistic basis for interpreting PFAS adsorption and desorption on MPs surfaces under environmentally relevant conditions.

One primary mechanism governing interaction between MPs and PFAS is hydrophobic interactions. Hydrophobic interactions arise because water repels nonpolar molecules, effectively forcing molecules with nonpolar functional groups together while structuring surrounding water molecules more tightly. Since most environmental MPs are inherently nonpolar, this interaction is a dominant mechanism driving contaminant adsorption [75]. The strength of this process depends on both the contaminant and polymer characteristics [76]. For instance, Yang et al. [77] showed that polystyrene pellets exposed to Fenton’s reagent for nine days experienced a reduction in contact angle from 127.75° to 99.81°. This reduction is partly due to the introduction of oxygen-containing surface groups, which increase polarity [59]. Longer carbon chains usually enhance hydrophobicity and thus favor sorption to MPs [78]. However, differences in polarity can modify this trend. For example, although PFOA and perfluorooctane sulfonamide (PFOSA) share the same chain length, the sulfonate group of PFOA increases polarity, resulting in lower adsorption compared with PFOSA on polyethylene particles [6]. Hydrophobic affinity-driven partitioning is therefore most frequently reported as the dominant interaction mechanism for nonpolar polymers such as polyethylene and polypropylene under environmentally relevant conditions (Table 1).

Table 1. Dominant interaction mechanisms governing PFAS sorption to MPs under environmental conditions.

Polymer Type	Dominant Interaction(s)	Key Environmental Modifiers	Implications for Environmental Fate	References
PE, PP	Hydrophobic affinity driven partitioning	Aging, salinity	Promotes long-term PFAS association with buoyant MPs, facilitating horizontal transport across surface waters and enhancing redistribution from source regions to remote marine environments.	[6,78]
PS	Electrostatic + π interactions	pH, ionic strength	Results in condition-dependent PFAS binding that may shift between sequestration and release, leading to variable bioavailability during transitions between freshwater, estuarine, and marine systems.	[79]
PA	Pore filling	pH, PZC, DOM	Favors retention of PFAS within polymer matrices, potentially acting as a delayed-release reservoir that prolongs environmental persistence and exposure duration in sediments and soils.	[80]

Note: Table 1 provides a qualitative synthesis of dominant physicochemical interaction mechanisms reported to govern sorption between MPs and PFAS under specific experimental and environmental conditions. Interaction dominance reflects the physicochemical context of individual studies, including polymer type, PFAS class, solution chemistry, and analytical approach, rather than a universal ranking applicable across all systems. Accordingly, Table 1 does not include quantitative sorption coefficients, kinetic parameters, or polymer-specific uptake rates, and should not be interpreted as providing predictive or comparative metrics of PFAS sorption behavior.

Another mechanism of interaction between MPs and PFAS includes electrostatic interactions. Functional groups may deprotonate as pH increases, leading to changes in surface charge state (e.g., neutral to negative or positive to neutral), which in turn modulate electrostatic interactions with PFAS [81]. Under acidic conditions, protonation patterns can create attractive charge differences that promote PFAS binding to MPs [82–84]. For instance, Wang et al. [6] observed that PFOS adsorption onto PE and PS increased as solution pH decreased due to protonation of the polymer surfaces under acidic conditions, with increasingly positive charges favoring the binding of the negatively charged PFOS molecules. Consistent with these observations, electrostatic interactions are most often identified as dominant for polystyrene under pH and ionic strength–dependent conditions, as summarized in Table 1.

Shifts in pH alter the ionization states of both PFAS and MPs surfaces, thereby changing surface charge and altering electrostatic interactions [85]. When the environment becomes more alkaline, however, the resulting like-charge configuration tends to weaken sorption forces [85–88]. Empirical results by Mejías et al. [85] showed that lowering the pH from 7 to 4 enhanced PFAS uptake by PA MPs, whereas adsorption dropped once the pH rose beyond the polymer’s point of zero charge (pHPZC) because the negatively charged PA surface repelled anionic PFAS molecules [80]. Competing solutes such as dissolved ions or organic matter further complicate this equilibrium, and their own behavior is likewise sensitive to pH, underscoring the central importance of solution chemistry in regulating PFAS and MPs interactions [85,89,90].

Electrostatic interactions are also affected by ionic strength. In the same study, the addition of sodium and calcium ions enhanced PFOS sorption to PE, demonstrating how cations can strengthen charge-based associations between PFAS and MPs. Elevated ionic strength can suppress PFAS sorption onto MPs through competitive interactions with co-occurring ions; in line with this, Mejías et al. [85] observed a marked decrease in PFOS uptake by PA in a saline matrix [85], while NaCl had little effect on several other compounds. High DOM concentration generally reduces PFAS sorption to MPs via site competition and complexation with hydrophobic moieties of humic/fulvic substances [91–93]. The

dampening effect is typically stronger for long-chain PFAS (e.g., PFOS and PFOA) and weaker for short-chain analogs such as perfluorobutanoic acid and perfluoropentanoic acid (PFBuA and PFPeA). For instance, raising humic acid to 25 mg L⁻¹ lowered adsorption on PA from 92% to 26% for PFOS and from 20% to 3.2% for PFOA, whereas short-chain PFAS were less affected. However, complexation of PFAS with humic acid was shown to enhance PFAS removal by coagulation, and this effect is being enhanced for water treatment [79,94,95]. These matrix-dependent effects, including ionic strength and dissolved organic matter, are reflected in the dominant interaction modifiers reported across polymers in Table 1.

A third mechanism of interaction between MPs and PFAS is pore-filling, the occupation of micropores and nanopores within polymer matrices by contaminant molecules at the limits of their solubility, similar to adsorption [76,95]. Among the various ways contaminants attach to MPs, the pore-filling process stands out, especially in compact or glassy polymers such as polyamide (PA), PVC, and PS [96]. In these materials, pollutant molecules do not simply stick to the surface; instead, they slowly move inward, settling into the tiny pores that run through the plastic matrix. How efficiently this happens depends largely on the size match between the pores and the contaminant molecules. The bigger voids tend to be occupied first, while diffusion slows once adsorption begins to reach the smaller cavities [76]. Weathering processes can intensify this pathway by creating additional surface pores, thereby expanding potential sorption sites [97]. Evidence for pore-filling has been documented in interactions between MPs and various pollutants, including ciprofloxacin [98], dichloro-diphenyl-trichloroethane (DDT) [90], and heterocyclic compounds [99]. For PFAS, though, clear proof of the same process is still scarce. The most convincing data so far come from Mejías et al. [85], who examined PA particles with scanning electron microscopy. Before exposure, the polymer showed many open pores, but after contact with PFAS, these openings appeared blurred or sealed, suggesting that the compounds had either filled or blocked them. Accordingly, pore-filling is most consistently reported for compact or glassy polymers such as polyamide, where this mechanism emerges as dominant under specific pH and DOM conditions (Table 1).

Multiple matrix-dependent forces contribute to PFAS affinity for MPs, including specific intermolecular interactions, electrostatic attraction/repulsion, hydrophobic partitioning, salting out, cation bridging, and water cluster effects [100–103]. Two consistent patterns emerge from prior work: (i) ionized PFAS can adsorb more strongly to PS in seawater than in freshwater, attributed to salting out and cation bridging, whereas neutral FOSA shows greater adsorption in freshwater [79], and (ii) in seawater, hydrophobic interactions are dampened to the extent that adsorption coefficients no longer increase systematically with PFAS chain length, unlike in freshwater systems [104–106]. Rather than reflecting separate or additive mechanisms, these observations indicate that hydrophobic, electrostatic, and matrix-dependent interactions operate concurrently, with their relative importance shifting as a function of polymer properties and environmental context. To facilitate synthesis across a diverse and rapidly expanding literature, Table 1 organizes reported MPs–PFAS interaction mechanisms according to dominant physicochemical processes and the environmental or material conditions under which they have been observed. Collectively, these recurring mechanistic patterns and controlling factors across polymer types are synthesized in Table 1.

To facilitate synthesis across a diverse and rapidly expanding literature, Table 1 organizes reported MPs–PFAS interaction mechanisms according to dominant physicochemical processes and the environmental or material conditions under which they have been observed, synthesizing recurring mechanistic patterns and controlling factors across polymer types.

A study of PFAS adsorption on granular activated carbon (GAC) showed that long-chain PFAS adsorption was not impacted by temperature but that short-chain PFAS had an up to fourfold decrease in adsorption with a temperature increase from 10 to 30 °C [107]. The effect of temperature on adsorption of PFAS onto MPs merits similar study.

Interactions and Combined Effects

Section 3 synthesizes and extends the interaction mechanisms described above by explicitly linking physicochemical adsorption processes with observed biological effects under co-exposure conditions. Building on the interaction mechanisms described above, this section focuses on how these physicochemical processes translate into combined biological and toxicological effects. Persistent environmental chemicals such as PFAS and EDCs readily associate with MPs surfaces in a wide range of environmental conditions. The hydrophobic character and extensive surface area of these particles make them particularly effective at accumulating pollutants including polybrominated diphenyl ethers (PBDEs), pharmaceuticals, EDCs, and other persistent organic pollutants (POPs) present in the water column [108]. The attachment and subsequent release of these compounds from MPs are governed by complex physicochemical interactions that depend on multiple variables. Key controlling factors include the polymer's composition, morphology, and surface energy, as well as its roughness and porosity, all of which determine how easily contaminants can adsorb or desorb. Environmental influences, such as pH, salinity, temperature, and organic matter content, further modify these processes. Equally important are the intrinsic properties of the pollutants themselves: solubility, charge, redox activity, and persistence all shape their affinity for plastic surfaces [109,110]. Parameters like the octanol–water partition coefficient (K_{ow}) and the degree of polymer weathering are especially decisive in adsorption dynamics, dictating whether contaminants remain bound or are released back into the environment. As a result, different polymers display distinct behaviors in contaminant interactions. PE, PP, PS, and PVC each demonstrate unique sorption capacities and desorption tendencies, reflecting variations in their surface chemistry and aging characteristics [79,111–113].

Given that MPs and PFAS often co-occur in the environment and even in consumer products, scientists are increasingly studying their combined effects. There are multiple ways in which MPs and PFAS interactions can impact toxicity: MPs can adsorb PFAS from the environment, potentially acting as a transport vector, and organisms may be exposed to MPs and PFAS simultaneously, whether bound together or not. Moreover, products like PFAS-coated textiles or non-stick cookware directly release both MPs fibers/particles and PFAS into the environment [11]. Understanding whether co-exposure exacerbates or modifies toxicity is therefore essential for accurate environmental risk assessment.

5. Combined Exposure and Toxicological Effects

Recent research suggests a synergistic relationship: the toxicity of MPs and PFAS together can be greater than either alone [2,11]. A primary reason is that MPs can increase the bioavailability and uptake of PFAS in organisms [11]. This vector-mediated uptake mechanism is consistently documented across taxa and exposure media, as summarized in Table 2 below, which synthesizes controlled laboratory and in vitro studies examining combined MPs–PFAS exposure. MPs can sorb PFAS from water, and when an organism ingests those particles, the attached PFAS are delivered at a higher local concentration to the gastrointestinal tract. MPs themselves, especially if small, may cross cell barriers and carry PFAS into tissues. This facilitates PFAS accumulating to higher levels inside the organism than would occur from PFAS exposure alone [11]. For instance, Sobhani et al. [10], examined how exposure to PVC MPs influences the bioaccumulation of PFOS and PFOA

in earthworms, as well as their reproductive performance. The results showed that the bioaccumulation factors (BAFs) for both PFOS and PFOA increased by as much as 2.5-fold when earthworms were maintained in soils containing MPs. At concentrations of 500 and 1000 mg kg^{−1} soil, PVC MPs promoted increased PFAS uptake and led to a marked decline in reproductive output.

Table 2. Studies on combined MPs and PFAS exposure.

Study (Organism)	Co-Exposure Design	Key Findings on Interaction	Reference
Zebrafish	Fish fed with MPs alone, PFAS (PFOS/PFOA) alone, or PFAS pre-adsorbed onto MPs	Combined exposure (PFAS-coated MPs) caused significantly greater toxicity than individual exposures. Fish showed disrupted organ homeostasis and liver stress markers far worse in the combined group, indicating synergistic harm to organ systems.	[114]
Earthworms	Soil invertebrates in soil spiked with PFOA/PFOS, with and without added MPs fragments	Presence of MPs in soil increased PFAS bioaccumulation ~2.5× in earthworms compared with PFAS without MPs. MPs sorbed and ferried PFAS into worm tissues, leading to higher internal doses and enhanced toxic effects on worm growth and survival.	[10]
<i>Daphnia magna</i>	Freshwater planktonic crustaceans exposed to MPs + PFAS (PFOA) at varying concentration ratios	Interaction effects varied with dose ratio: At some ratios, MPs and PFOA acted synergistically to reduce <i>Daphnia</i> feeding and reproduction (toxicity greater than sum of parts), while at other ratios effects were merely additive or slightly antagonistic. This shows that outcome of MPs–PFAS co-exposure can depend on relative concentrations, possibly due to limited adsorption capacity or biological compensatory responses.	[110]
Human intestinal cells	Cells were exposed to PFOS alone, PS-MPs alone, and combined PFOS + PS-MPs treatments	Low-dose PFOS exposure with polystyrene MPs (PS-MPs) resulted in a mitigating or “cleaning effect.” The PS-MPs adsorbed part of the PFOS, reducing its bioavailability and cytotoxicity to intestinal cells. Desorption at higher concentrations could later increase PFOS uptake and toxicity.	[115]
Human intestinal cells	Exposed intestinal epithelial cells to PFAS alone, MPs alone, and combinations thereof (i.e., PFAS pre-adsorbed onto MPs or co-incubated) across a gradient of concentrations	Co-exposure of human intestinal epithelial cells to PFAS and MPs altered PFAS uptake and toxicity in a dose- and particle-dependent manner: MPs could sorb PFAS, reducing its free concentration and cellular toxicity under certain conditions, but also facilitate PFAS internalization or desorption at higher doses, thereby modulating net cell exposure.	[116]
Human intestinal cells	In vitro co-exposure of human intestinal Caco-2 cells to PET MPs plus PFOA	PFOA alone induced oxidative stress, mitochondrial dysfunction, and reduced expression of tight-junction proteins. In Caco-2 cells, compromised intestinal barrier integrity and when co-exposed with PET MPs, these harmful effects were exacerbated via increased PFOA accumulation (through reduced membrane permeability) and stronger inhibition of tight-junction proteins.	[117]

Note: Table 2 represents a qualitative synthesis of dominant interaction mechanisms reported across multiple experimental and observational studies. Reported mechanisms may co-occur and vary depending on polymer type, polymer aging state, PFAS chain length, particle size and environmental context (e.g., exposure medium, concentration ratios).

Laboratory aquatic studies mirror this finding. Rainieri et al. [114], exposed zebrafish to MPs, PFAS, or a combination (PFAS sorbed on MPs). Fish receiving the combined exposure had more severe toxic effects, notably, greater liver damage and disruption of organ homeostasis, than fish exposed to only MPs or only PFAS [11]. PFAS sorbed to MPs likely concentrated in the liver, overwhelming detoxification pathways. Likewise,

mussels exposed to MPs plus PFAS accumulated more PFAS in their tissues than those exposed to PFAS alone [11]. A study on clams found that smaller-sized MPs particles led to higher ingestion rates and subsequently higher PFOS accumulation in the clams, illustrating that MPs size can influence the degree of PFAS uptake [11]. In *Daphnia magna* (water fleas), combined exposure experiments revealed a complex dose-ratio-dependent interaction: at certain concentration ratios, MPs and PFAS showed synergistic toxicity (enhancing each other's lethal or sublethal effects), while at other ratios the interaction was only additive (effects similar to sum of individual effects) or even antagonistic (reduced effects with one mitigating the effects of the other) due to higher levels of MPs causing premature gut fullness [118]. This suggests that while synergy is often observed, it may vary with conditions and concentrations affecting saturation of sorption sites on MPs or biological stress thresholds. As reflected in Table 2, these dose-ratio-dependent outcomes demonstrate that MPs–PFAS co-exposure effects are not uniformly synergistic, but instead vary with relative concentrations, sorption site availability, and organismal feeding or uptake dynamics.

In an experiment conducted by Jian et al. [15], adult female zebrafish were subjected for 14 days to either 10 µg/L PFOS (or its substitute, F-53B), 50 µg/L MPs, or mixtures of both to assess individual and combined toxic effects. The addition of MPs lowered the freely dissolved concentrations of PFOS and F-53B in the water but did not alter their accumulation in hepatic or intestinal tissues. When zebrafish experienced joint exposure, the most pronounced responses were observed in the liver, including heightened oxidative stress, activation of immunoinflammatory pathways, and disruption of energy metabolism. Sequencing of 16S rRNA genes indicated that F-53B together with MPs produced the strongest alterations in gut microbial communities. Functional predictions suggested these microbial shifts could impair pathways linked to immunity and energy regulation. Furthermore, significant associations emerged between microbial community changes and host immune/metabolic indicators. Collectively, the findings show that co-exposure to PFOS/F-53B and MPs intensifies liver immunotoxicity and metabolic disturbances in zebrafish, most likely mediated by gut microbiota dysregulation, compared with single-contaminant exposures.

Research conducted by Soltanighias [119], explored the long-term toxicological impacts of PFAS and MPs on *Daphnia*, a keystone genus in freshwater ecosystems and a widely used ecotoxicological model. The study assessed the effects of PFOS, PFOA, and polyethylene terephthalate (PET) MPs, both individually and in combination, on multiple ecological endpoints. To move beyond traditional single-genotype assays, two *Daphnia* genotypes with contrasting histories of chemical exposure were included. The results demonstrated that PFAS and PET MPs induced developmental abnormalities, delayed reproductive maturity, and inhibited somatic growth. Prior exposure to polluted environments reduced the tolerance of one genotype, suggesting cumulative fitness trade-offs. Mixture analyses indicated that joint effects were predominantly additive (59%) but also frequently synergistic (41%), with no evidence of antagonism. Genotype-specific variability underscores how genetic background and exposure history shape organismal sensitivity, highlighting the need to integrate multiple genotypes into environmental risk assessments for more reliable predictions of ecological consequences. Overall, studies summarized in Table 2 demonstrate consistent patterns across taxa: MPs increase PFAS bioavailability by acting as sorptive and transport vectors; co-exposure most often produces additive or synergistic toxicity; and outcomes depend strongly on particle size, polymer type, and exposure ratio. Table 2 indicates that the magnitude and direction of these effects are context dependent, with exposure medium, dose ratios, and particle aging influencing whether MPs–PFAS interactions are additive, synergistic, or occasionally antagonistic. Evidence shows that

these two contaminants do not accumulate in the same tissues. In fish, MPs are primarily retained in the gills and intestines, whereas PFAS tend to concentrate on systemic compartments such as blood, muscle, liver, and reproductive tissues. Regarding biological effects, both pollutants are linked to growth suppression, behavioral disturbances, neurological impairment, and molecular-level toxicity. However, MPs more distinctly affect digestive processes and the gut microbial community, while PFAS exposure is strongly associated with impacts on the reproductive system, including alterations in sex ratios. Taken together, simultaneous exposure to MPs and PFAS may produce toxic outcomes that differ in nature and severity compared with PFAS exposure alone [12]. Evidence summarized in Table 2 further indicates that such modified outcomes extend to human-relevant in vitro intestinal models, underscoring the potential implications of MPs–PFAS co-exposure for internal dose modulation and risk assessment. Figure 2 represents conserved toxicological effects of MPs and PFAS across aquatic and terrestrial organisms, focusing on the behavior and nervous system, digestive system, growth and inhibition, immune system, metabolism and oxidative stress, sex and reproductive system, and molecular-level genes and proteins [12].

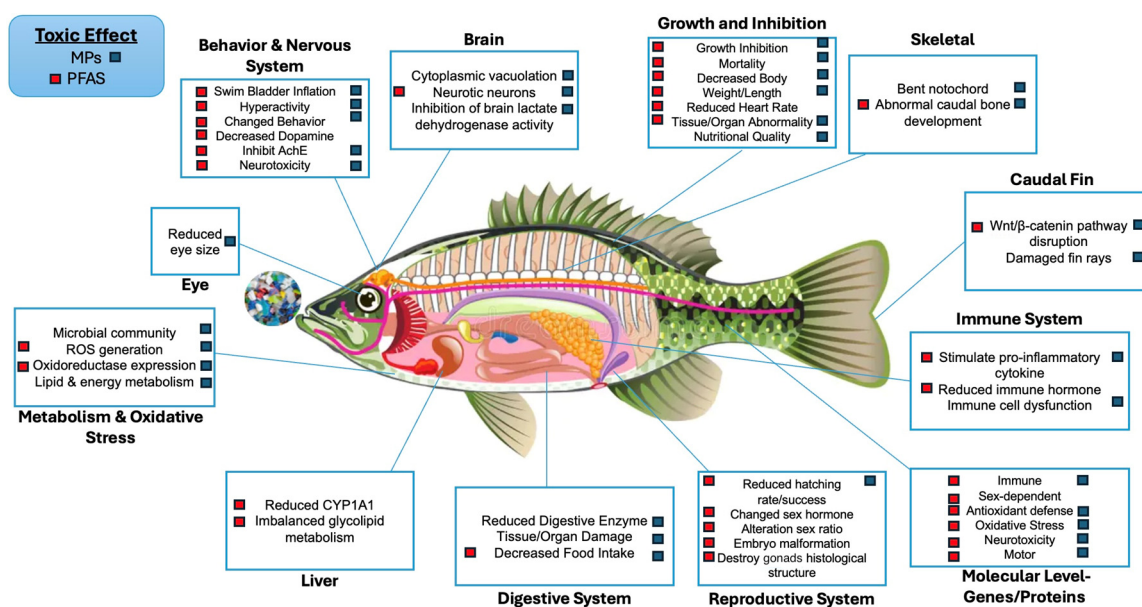


Figure 2. Conserved toxicological effects of MPs and PFAS across aquatic and terrestrial organisms. The illustrated pathways, neurological disruption, digestive impairment, growth inhibition, immune dysfunction, oxidative stress, reproductive toxicity, and molecular-level effects that are applicable not only to fish but also to bivalves and mammals. Adapted from Dai et al. [12] and Chen et al. [19] under a Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/> accessed on 23 September 2025).

A study conducted by Zou et al. [117], examined the combined impact of PET MPs and PFOA on human intestinal cells using an in vitro model, focusing on both cytotoxicity and barrier integrity. PFOA exposure alone triggered oxidative stress, impaired mitochondrial function, and downregulated tight junction (TJ) proteins, leading to disruption of the intestinal barrier. When PET MPs were present, these adverse effects were intensified: MPs reduced membrane selectivity, facilitated greater intracellular accumulation of PFOA, and amplified its inhibitory effect on the tight junction protein zonula occludens-1 (ZO-1).

Human exposure to PFAS and micro- and nano-plastics (MNPs) presents substantial health concerns as these pollutants can interfere with multiple organ systems, disrupt metabolic functions, and impair elimination pathways. Both have been associated with heightened oxidative stress, inflammatory responses, and widespread disturbances at the cellular and physiological levels. When occurring together, PFAS and MNPs raise

additional concern due to their potential to exacerbate liver injury, kidney dysfunction, endocrine imbalances, and harmful effects on reproduction and development [120].

The studies above illustrate that the interplay between MPs and PFAS is complex. Generally, the co-exposure heightens concern because organisms do not encounter these stressors in isolation in the real world. MPs and PFAS are often present together in polluted environments (e.g., wastewater effluent contains both [121]). As a result, risk assessments that consider only one pollutant at a time might underestimate real-world toxicity. The heightened effects (synergy) are largely attributed to mechanisms like increased uptake (MPs carrying PFAS across biological barriers) and combined burden on detox systems (e.g., liver dealing with both particle-induced inflammation and PFAS-induced metabolic disruption). That said, as noted in the intestinal cell study, there can be scenarios where the presence of one mitigates the other's impact; for example, a MPs particle binding a molecule of PFAS could render that PFAS temporarily inactive biologically. This "cleaning" or antagonistic effect might occur under limited conditions (such as low pollutant levels and short exposure times) [11]. However, because MPs will eventually release the pollutant (adsorption is reversible), it may simply be delivered to a different location or at a later time. Therefore, any short-term antagonism could turn into delayed exposure. From a toxicological mechanism's standpoint, combined exposure means multiple pathways are engaged, which includes physical stress and inflammation from MPs and chemical toxicity from PFAS. One example is the zebrafish result where liver antioxidant systems were likely overwhelmed by having to cope with both oxidative stress from MP's presence and PFAS's hepatotoxic effects, leading to greater cellular damage. Another example is immune function in that PFAS are immunotoxins, so an organism exposed to PFAS might not mount a proper immune response to clear inhaled MPs debris from the lungs, resulting in more particle-induced lung injury than would occur in the absence of PFAS. In the context of carcinogenesis, MPs may create the oxidative stress and chronic inflammation that promote cancer, while also transporting PFAS and other toxins that further trigger tumor progression [54]. In essence, synergistic ecotoxicological effects of MPs and PFAS represent a growing concern, and researchers emphasize the need to study such interactions more deeply [11].

6. Environmental and Health Implications

Mixture toxicity is often worse than either stressor alone. Across zooplankton and early-life stages, co-exposure tends to depress growth, delay maturation, and reduce fecundity beyond additivity which is consistent with mixture/carrier effects. Recent work with *Daphnia* shows developmental failure, delayed sexual maturity, and stunted growth under MPs and PFAS versus single exposures [122]. PFAS adsorb to common polymers (PET, PE, PP, and PVC) via electrostatic/hydrophobic interactions—modulated by pH, NOM, ionic strength, and temperature—which enhances co-transport and organism exposure when particles are ingested [74]. MPs with sorbed PFAS can be ingested by plankton and other invertebrates and transferred up the food web, with reported intestinal damage, inflammation, liver, and oxidative stress endpoints in aquatic biota (with PFAS burden as a key driver) [123]. Co-occurrence in air and wet/dry deposition (rain and aerosols) can deliver MPs and PFAS mixtures to surface waters; meteorology (UV, wind, and humidity) shapes loading and timing [124].

MPs in soil can increase PFOS/PFOA bioaccumulation and suppress reproduction in earthworms, indicating MPs can act as PFAS vectors in detrital food webs, which is relevant for nutrient cycling and top-down control. New kinetic data also describes PFAS uptake/elimination dynamics in worms [10]. MPs alter microbial diversity, network stability, and carbon/nutrient processes; emerging work highlights distinctive microbiomes in MPs

contaminated soils, which likely feed back into PFAS/MPs degradation and sorption behaviors [124]. The same vectoring processes that elevate PFAS exposure in aquatic/terrestrial organisms imply higher PFAS transfer risk to humans via seafood, irrigation pathways, dust, and drinking water precursors, although quantification of incremental risk from MP-sorbed PFAS compared with PFAS alone is still developing. State-of-the-science reviews call this a priority research gap [47].

Despite growing evidence for MPs–PFAS co-occurrence and vectoring, quantitative attribution of health outcomes to particle-bound versus dissolved PFAS remains limited [11,45]. Most toxicological and epidemiological studies do not distinguish exposure pathways, aging state of MPs, or desorption kinetics within biological systems, constraining dose–response interpretation and mixture risk assessment [45–47]. Addressing these uncertainties will require harmonized co-exposure study designs that explicitly track PFAS partitioning between dissolved and MPs-associated fractions across environmental and biological compartments [43,125].

Seafood is the dominant dietary PFAS source, especially shellfish and some marine/freshwater fish. Recent field and dietary studies (including New England cohorts and market surveys) consistently find the highest PFAS in shrimp and lobster and variable levels in finfish [126]. Humans ingest MPs particles mainly via bivalves and small whole-edible seafood (oysters, mussels, and small shrimp), where the gut is consumed. Typical seafood MP burdens span ~0.2–4 particles/g for bivalves (values vary widely by study and method) [127].

PFAS can bioaccumulate and biomagnify. Many PFAS (notably PFOS and some ether sulfonates) show trophic magnification in aquatic food webs and recent syntheses, and models confirm biomagnification and tissue partitioning mechanisms. Recent reviews emphasize reduced vaccine responses as the most consistent human finding [128]. With the same vectoring in human intestinal cell models, PET MPs and PFOA co-exposure increases oxidative stress and tight-junction/barrier disruption beyond single-agent exposures, which mechanistically enhances passage of chemicals or particles under some conditions. Animal work shows MPs can impair gut integrity and alter immune signaling, which may further enhance PFAS uptake and internal dose under co-exposure conditions [118].

Current law controls PFAS mixtures in water and food and MPs emissions at the source, but no regulation directly evaluates or limits PFAS carried by MPs. Near-term fixes are procedural: explicitly monitor PFAS on MPs fractions, apply mixture tools (hazard index (HI), mixture assessment factor (MAF)) to particle-bound chemicals, and scale source controls (tires, textiles, biosolids, and packaging). In the medium term, both the EU REACH PFAS restriction and the Global Plastics Treaty are the natural places to formalize the MPs and PFAS interaction risk [129]. One policy/regulation suggestion is to use mixture tools but extend them to particle-bound PFAS. The U.S. EPA’s PFAS HI maximum contaminant level (MCL) for perfluorohexane sulfonic acid perfluorobutanesulfonic acid (PFHxS) and perfluorononanoic acid (PFNA), hexafluoropropylene HFPO-DA, and PFBS is an enforceable mixtures instrument, and it can be adapted to include a co-exposure term for PFAS sorbed to MPs when measured above a de minimis level. EFSA’s combined exposure guidance and EU work on an MAF give parallel anchors on the EU side [72]. Another suggested policy/regulation is to require co-monitoring and co-reporting in water using approved MPs methods. The EU just adopted a harmonized methodology to measure MPs in drinking water under the Drinking Water Directive; California already has a two-phase MPs monitoring playbook. Both can be amended to extract the MPs fraction and analyze PFAS on that fraction (liquid chromatography–mass spectrometry/mass spectrometry (LC-MS/MS)) and then co-report dissolved PFAS (µg/L) alongside MPs-bound PFAS (ng

PFAS per mg MPs or per particle) [130]. Another policy/regulation suggestion is to tackle high-emitting MPs sources that also emit PFAS.

New legal frameworks to limit tire abrasion (Euro 7) and the United Nations Economic Commission for Europe (UNECE) methodology to quantify tire particle emissions create a lever to cut a dominant MPs source that can carry PFAS from roads to waters. One approach is to build PFAS on tire and road [71] wear particle (TRWP) checks into the type of approval and market surveillance data streams. Another suggestion is to bridge food rules with particle monitoring. The EU's PFAS maximum levels in foods can explicitly ask labs to note MPs occurrence and, where feasible, quantify PFAS associated with isolated MPs fractions in whole-edible species (bivalves and small shrimp). That would transform current food PFAS compliance checks into interaction-aware surveillance [71]. As summarized in Table 3 below, the current rules manage PFAS mixtures (e.g., EPA HI; EU food MLs) and reduce MPs emissions (e.g., REACH 2023/2055; Euro 7) separately. None of these instruments define, measure, or regulate PFAS on MPs as a distinct exposure compartment, even though harmonized MPs methods now exist, and mixture frameworks can incorporate an extra pathway. These proposals simply connect the dots using legal tools that are already in force [131].

Table 3. Suggested interaction-focused regulatory actions for MPs-bound PFAS.

Action	Regulatory Program/Domain	What Is Required (Interaction Focused)	Reference
Define and report MPs-bound PFAS compartment	Water, wastewater, biosolids, seafood	Paired results: (a) dissolved PFAS and (b) PFAS on isolated MPs; include gut bioaccessibility (desorbable fraction)	[132]
Extend mixture metrics to include particle-bound dose	Drinking water (MCL/HI), food (EFSA/MAF)	Add site-specific MPs-bound PFAS term to EPA HI (as dissolved equivalent via gut desorption); apply MAF/combined exposure with explicit particle-borne pathway	[71]
Hot spot triggers for interaction risk	Ambient water, shellfish sanitation, CERCLA, stormwater	Mandate MPs × PFAS co-monitoring at predefined hot spots	[72]
Wastewater and biosolids: measure the complex	NPDES/WWTP permits; biosolids	Quantify PFAS on sludge-borne MPs and in effluent MPs fractions	[73]
Seafood advisories that are interaction aware	Fish/shellfish advisories; food safety	Flag elevated MPs occurrence and, where available, MPs-bound PFAS alongside PFAS-in-tissue data	[71]
Mobility: Euro 7 tire-wear + PFAS screens on TRWP	Vehicle type approval; non-exhaust emissions	Implement tire abrasion limits and include PFAS screens on TRWP in conformity of production datasets	[133]
Source restrictions addressing the pair	REACH/market restrictions	Enforce restriction on intentionally added MPs (2023/2055); require substitution plans where PFAS-containing polymer additives could create PFAS-laden MPs	[134]

Note: Table 3 summarizes the suggested interaction-focused regulatory actions for MPs-bound PFAS.

7. Research Gaps and Future Directions

Despite rapid advances in understanding MPs and PFAS individually, critical gaps persist in evaluating their combined environmental behavior, exposure pathways, and toxicological impacts. The most fundamental limitation is the lack of harmonized methodologies capable of capturing MPs–PFAS interactions across sampling, analytical, biological, and modeling domains. Beyond methodological inconsistency, several conceptual gaps limit interpretation of MPs–PFAS interactions. Most studies isolate single mechanisms (e.g., sorption or toxicity) without integrating feedback among particle aging, biofilm formation, desorption kinetics, and organismal uptake, despite strong evidence that these processes evolve dynamically over time [23,40,42,43]. In addition, mixture toxicity frameworks rarely distinguish between dissolved and particle-bound PFAS exposure, limiting causal

attribution of observed biological effects and internal dose pathways [11,43]. Population-level sensitivity; genotype-dependent responses; and long-term, low-dose co-exposure scenarios remain poorly characterized, particularly for terrestrial food webs and human-relevant endpoints, where extrapolation from short-term laboratory studies remains highly uncertain [16,40,121].

7.1. Monitoring, Sampling, and Analytical Standardization

A major gap lies in inconsistent sampling strategies and quality assurance practices. Laboratories vary widely in the use of sampling materials, procedural blanks, PFAS-free consumables, and contamination controls, leading to substantial inter-study variability. These discrepancies are particularly problematic for MPs–PFAS interaction research, where background contamination from fluoropolymer labware can severely bias results. Harmonized QA/QC frameworks that integrate PFAS-safe practices into established MPs monitoring protocols are therefore urgently required [135].

Particle isolation and identification further limit comparability. Differences in pre-treatment methods (oxidative vs. enzymatic digestion), size cut-offs, and analytical instrumentation (μ FTIR, μ Raman, and TD-GC/MS) yield non-equivalent particle counts and polymer classifications. Recently implemented harmonized methodologies have been adopted, such as the EU Drinking Water Directive MPs protocol and California’s phased monitoring framework, which provides a realistic template for standardized MPs characterization across environmental matrices [71].

7.2. Reporting Metrics and Interaction-Relevant Data Gaps

Another major limitation is the lack of standardized reporting units for MPs–PFAS interaction studies. Sorption is variously expressed as mass-based (ng PFAS mg^{-1} MPs), particle-based, percentage removal, or equilibrium coefficients without consistent reporting of particle surface area, morphology, or size distribution. This prevents meaningful synthesis across studies. Future research should report, at minimum, (i) particle number, mass, and surface area distributions; (ii) polymer and morphology; (iii) K_d/K_f plus test conditions (pH, ionic strength, DOM, and temperature); and (iv) optional area-normalized sorption (e.g., ng/cm^2). Research should use guidance on selecting informative metrics [11].

Equally important is harmonization of desorption and bioaccessibility assays. Current gut simulation protocols vary widely in composition and contact times, limiting interpretation of dietary exposure risk. Adoption of standardized INFOGEST-based digestion models for quantifying PFAS release from MPs would enable more realistic estimates of internal dose and trophic transfer potential [43].

7.3. Environmental Realism: Aging, Biofilms, and Mixture Complexity

Most experimental studies continue to rely on pristine MPs, despite strong evidence that environmental aging and biofilm colonization fundamentally alter surface chemistry, particle density, and contaminant affinity. Weathering-induced changes in roughness, polarity, and functional group abundance strongly influence PFAS sorption/desorption behavior. Future studies should apply tiered artificial aging and biofilm protocols with quantifiable surface-state metrics (e.g., carbonyl index, roughness, and biofilm coverage) reported alongside interaction results to improve environmental relevance [42].

7.4. Modeling, Transport, and Ecological Integration

Current mass balance and fate models largely treat PFAS as dissolved-phase contaminants and MPs as inert particles. This separation fails to capture time-dependent sorption kinetics, desorption delays, and episodic transport associated with stormwater and shoreline processes. Future modeling efforts should incorporate kinetic sorption–desorption

parameters, polymer-specific capacities, and environmentally responsive surface-state variables that evolve with aging and biofouling [74].

At the ecosystem scale, coupled watershed–estuary–coastal models that explicitly include MPs beaching, remobilization, and PFAS load transfer are needed to resolve exposure hot spots and food web implications [136,137]. Integration of such models with toxicological thresholds and species sensitivity distributions would significantly advance risk-based environmental assessment.

8. Conclusions

Confronting this issue demands a coordinated, long-term strategy. Monitoring programs should capture plastics, dissolved PFAS, sediments, and organisms simultaneously, with particular attention to episodic discharges such as stormwater and combined sewer overflows. Reliable cross-study comparisons depend on standardized laboratory protocols that pair chromatographic and mass-spectrometric analyses of PFAS with spectroscopic or thermoanalytical identification of MPs. Equally important is the creation of interoperable data systems that connect ecological, toxicological, and regulatory information across geographic scales.

Despite growing evidence, major uncertainties persist as methods for quantifying sorption and desorption remain inconsistent, reporting units vary widely, and most experiments still rely on pristine plastics rather than weathered or biofilm-coated materials that better represent real environments. Understanding combined toxicity also requires attention to genetic diversity, population-level effects, and advanced modeling of mixture behavior. Moving forward, researchers should emphasize kinetic sorption frameworks, dynamic biofilm parameters, and coupled hydrodynamic models that resolve shoreline and coastal exchange processes.

This review brings together evidence indicating that interactions between MPs and PFAS are not consistently synergistic or negligible but instead depend strongly on context. Differences in polymer composition, particle size, environmental aging, surrounding chemistry, and biological conditions all influence whether MPs amplify, delay, or modify PFAS exposure and effects. Earlier risk paradigms generally treated contaminant transfer via MPs as minor relative to dissolved-phase exposure; however, the studies synthesized here show that MPs-associated PFAS can meaningfully increase internal burdens in sensitive organisms, particularly during chronic exposure and in lower trophic levels, calling into question assumptions built into conventional exposure and fate models and pointing to the need for revised conceptualization of contaminant mixtures in natural systems.

The interaction mechanisms discussed here also carry important implications for environmental modeling and regulatory evaluation. Sorption and desorption of PFAS from MPs occur over time and introduce exposure patterns that are not captured by equilibrium-based fate and transport models. Evidence from gastrointestinal and bioaccessibility studies further indicates that particle-bound PFAS should be treated as a biologically available fraction rather than as permanently immobilized mass. Accounting for these dynamics in mixture-toxicity frameworks, hazard indices, and cumulative risk assessments will be critical to avoid underestimating exposure and effects in environments where MPs and PFAS co-occur.

Beyond environmental exposure, the combined presence of MPs and PFAS presents unique challenges for regulation and risk assessment. Current regulatory frameworks typically evaluate plastic pollution and chemical contaminants independently, despite mounting evidence that their interactions alter transport, bioavailability, and toxicity, limiting the ability of existing guidelines to account for mixture effects, delayed exposure through desorption, and cumulative organismal stress. Integrating MPs–PFAS interactions

into environmental quality standards, drinking water guidelines, and sediment assessment tools will be necessary to more accurately reflect real-world exposure scenarios. Such integration also requires closer alignment between monitoring programs and toxicological benchmarks, ensuring that mixture-specific effects are not systematically underestimated.

Ultimately, the convergence of MPs and PFAS defines a new frontier in environmental science. Their persistence and mutual reinforcement underscore the need for sustained monitoring, harmonized regulation, and genuine collaboration among scientists and policymakers alike. Without such measures, the ecological and human health consequences of these contaminants will continue to escalate. Inversely, proactive integration of science, monitoring, and policy offers a pathway to mitigate risks and protect both environmental integrity and public health.

This review advances the field by explicitly conceptualizing MPs as dynamic vectors that modify the environmental fate, biological availability, and toxicological consequences of PFAS rather than as inert co-occurring debris. By integrating physicochemical interaction mechanisms with organismal and cellular-level evidence, this synthesis demonstrates that MPs-mediated sorption and desorption can alter PFAS internal dose, tissue targeting, and mixture toxicity in ways not captured by single-contaminant frameworks. In particular, the evidence compiled here highlights how particle size, polymer chemistry, environmental aging, and biological conditions collectively govern whether MPs–PFAS interactions amplify; delay; or, in limited cases, temporarily mitigate toxic effects. Framing MPs–PFAS co-exposure as a coupled, time-dependent process provides a foundation for interaction-aware monitoring strategies, improved fate and transport modeling, and more realistic ecological and human health risk assessment.

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